

# Cloud Migration Assessment

Current state vs target state analysis

Document ID: EA-CLD-009

*This document is the property of ArchiSurance B.V. and contains confidential information. Unauthorized reproduction or distribution is prohibited. For questions contact the Enterprise Architecture team at [ea-team@archisurance.example.com](mailto:ea-team@archisurance.example.com).*

# Document Version History

| Ver. | Date       | Author          | Description                    |
|------|------------|-----------------|--------------------------------|
| 0.1  | 2024-01-15 | J. van den Berg | Initial draft                  |
| 0.5  | 2024-03-22 | M. de Vries     | Peer review incorporated       |
| 1.0  | 2024-06-10 | P. Jansen       | Approved by Architecture Board |
| 1.5  | 2024-11-04 | J. van den Berg | Updated for Q4 changes         |
| 2.0  | 2025-02-18 | M. de Vries     | Annual revision                |
| 2.1  | 2025-04-01 | P. Jansen       | Minor corrections              |

Table of Contents

1 Introduction ..... 1

2 Migration Drivers ..... 2

3 Current State Assessment ..... 3

4 Target State Architecture ..... 4

5 Application Migration Readiness ..... 5

6 Risk Analysis ..... 6

7 Migration Roadmap ..... 7

8 Appendices ..... 8

# 1. Introduction

This document assesses ArchiSurance's readiness for cloud migration and defines the target-state architecture. The migration is driven by the strategic goal to Migrate 100% Workloads to Cloud by 2027, championed by the CTO and the Enterprise Architect.

The assessment follows the '6 Rs' framework: Rehost, Replatform, Refactor, Repurchase, Retire, Retain. Each application component is evaluated against these options. For infrastructure details, see the Infrastructure Design Document (EA-INFRA-003).

A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Security review was completed in accordance with the CISO's security-by-design framework. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Change management activities include stakeholder communication, training sessions, and documentation updates. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. Stakeholders across multiple departments have contributed requirements for this workstream. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns.

Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog.

The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points.

Stakeholders across multiple departments have contributed requirements for this workstream. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner.

Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were

documented and tracked in the security risk register. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership.

## 2. Migration Drivers

The primary drivers are: (1) Reduce IT Operational Costs by 25% -- championed by the CFO, (2) Four-Nines Uptime Goal -- requiring Minimum 3 Availability Zones, and (3) the Legacy Oracle DB Migration Deadline Q4 2026 constraint.

Constraints: The Budget Cap EUR 2M Annual limits the pace of migration. The constraint 'No Public Cloud for PII Data (until 2026 review)' requires a hybrid approach for PII-handling systems. 'Vendor Lock-in Avoidance' mandates the use of cloud-agnostic technologies.

The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Lessons learned from the pilot phase have been incorporated into the current design iteration. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. Integration testing spans three environments: development, staging, and pre-production. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The cloud migration drivers has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations.

The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Security review was completed in accordance with the CISO's security-by-design framework.

External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The cloud migration cost drivers has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request.

Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The vendor selection process followed the procurement guidelines established by the CFO's office. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days.

### **3. Current State Assessment**

The current infrastructure consists of 10 compute nodes and 10 system software components. Key systems include PostgreSQL DB as the primary database, Oracle DB for legacy workloads, and K8s for container orchestration.

The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. The transition plan was developed in alignment

with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Integration testing spans three environments: development, staging, and pre-production. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The vendor selection process followed the procurement guidelines established by the CFO's office. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The current state infrastructure has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period.

In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline.

Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during

the rollout period. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations.

The vendor selection process followed the procurement guidelines established by the CFO's office. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice.

## 4. Target State Architecture

The target state leverages managed cloud services while maintaining the existing application architecture. Key changes include: migration of PostgreSQL to a managed database service, replacement of the on-premises Kubernetes cluster with managed K8s, and retirement of the Oracle database through data migration to PostgreSQL.

The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. Security review was completed in accordance with the CISO's security-by-design framework. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. Stakeholders across multiple departments have contributed requirements for this workstream. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. The architecture review confirmed that the proposed solution aligns

with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment.

Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period.

The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints.

The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days.

## 5. Application Migration Readiness

| Application            | Strategy   | Effort | PII? | Priority |
|------------------------|------------|--------|------|----------|
| CRM System             | Replatform | Low    | Yes  | P1       |
| Policy Administration  | Refactor   | Medium | Yes  | P2       |
| Claims Management Plat | Rehost     | High   | Yes  | P3       |
| Risk Engine            | Retain     | Low    | Yes  | P1       |

|                        |            |        |     |    |
|------------------------|------------|--------|-----|----|
| Customer Self-Service  | Repurchase | Medium | Yes | P2 |
| Document Management Sy | Replatform | High   | Yes | P3 |
| Notification Service   | Refactor   | Low    | Yes | P1 |
| Fraud Detection Engine | Rehost     | Medium | Yes | P2 |
| Reporting & Analytics  | Retain     | High   | No  | P3 |
| Enterprise Service Bus | Repurchase | Low    | No  | P1 |
| Identity & Access Mana | Replatform | Medium | No  | P2 |
| Billing System         | Refactor   | High   | No  | P3 |
| Broker Portal          | Rehost     | Low    | No  | P1 |
| Data Warehouse         | Retain     | Medium | No  | P2 |
| Workflow Engine        | Repurchase | High   | No  | P3 |
| API Gateway            | Replatform | Low    | No  | P1 |
| Content Delivery Netwo | Refactor   | Medium | No  | P2 |
| Email Service          | Rehost     | High   | No  | P3 |
| Mobile App Backend     | Retain     | Low    | No  | P1 |
| Batch Processing Engin | Repurchase | Medium | No  | P2 |

## 6. Risk Analysis

Key migration risks include data loss during cutover, performance degradation in shared cloud environments, regulatory non-compliance during transition, and skill gaps in the operations team. Mitigation strategies are defined in the Disaster Recovery Plan (EA-DR-010).

From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Stakeholders across multiple departments have contributed requirements for this workstream. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current

utilization rates. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Lessons learned from the pilot phase have been incorporated into the current design iteration. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024.

## 7. Migration Roadmap

| Phase   | Period     | Workloads              | Milestone                |
|---------|------------|------------------------|--------------------------|
| Phase 1 | Q1-Q2 2025 | Dev/Test environments  | Cloud landing zone ready |
| Phase 2 | Q3-Q4 2025 | Non-PII applications   | 5 apps migrated          |
| Phase 3 | Q1-Q2 2026 | PII apps (post review) | PII cloud approval       |
| Phase 4 | Q3-Q4 2026 | Oracle DB migration    | Oracle decommissioned    |
| Phase 5 | Q1-Q2 2027 | Remaining workloads    | DC exit complete         |

An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The cloud migration roadmap planning has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment.

An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer.

The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The

operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Change management activities include stakeholder communication, training sessions, and documentation updates. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline.

# 8. Appendices

## References and Related Documents

| Doc ID       | Title                               | Version |
|--------------|-------------------------------------|---------|
| EA-OVR-001   | Enterprise Architecture Overview    | 3.0     |
| EA-APP-002   | Application Portfolio & Inventory   | 4.2     |
| EA-INFRA-003 | Infrastructure Design Document      | 2.5     |
| EA-BIZ-004   | CRM Business Process Description    | 2.0     |
| EA-CLM-005   | Claims Processing Workflow          | 3.1     |
| EA-SEC-006   | Security Architecture               | 2.3     |
| EA-INT-007   | Integration Architecture            | 2.8     |
| EA-DAT-008   | Data Architecture & Governance      | 2.0     |
| EA-CLD-009   | Cloud Migration Assessment          | 1.5     |
| EA-DR-010    | Disaster Recovery & Business Contin | 3.0     |
| EA-NET-011   | Network Architecture                | 2.2     |
| EA-ROAD-012  | Strategic IT Roadmap 2025-2027      | 2.0     |
| EA-SVC-013   | IT Service Catalog                  | 3.5     |
| EA-VND-014   | Vendor & Technology Assessment      | 1.8     |
| EA-OPS-015   | Operational Runbook: Core Platform  | 4.1     |
| EA-CMP-016   | Compliance & Regulatory Requirement | 2.5     |
| EA-CHG-017   | Change Management: ERP Migration    | 1.2     |

## Distribution List

| Name            | Role                     | Responsibility      |
|-----------------|--------------------------|---------------------|
| J. van den Berg | Enterprise Architect     | Review & Approve    |
| M. de Vries     | Application Architect    | Review & Contribute |
| P. Jansen       | Integration Architect    | Review & Contribute |
| K. Bakker       | Infrastructure Architect | Review & Contribute |
| S. Visser       | Business Analyst         | Inform              |
| T. Mulder       | Process Architect        | Inform              |
| R. de Jong      | CISO                     | Review & Approve    |
| L. Hendriks     | Data Architect           | Inform              |
| A. Willems      | CTO                      | Final Approval      |
| B. Smit         | Head of Claims           | Business Sign-off   |

## Glossary of Acronyms

| Acronym | Definition                         |
|---------|------------------------------------|
| API     | Application Programming Interface  |
| BPM     | Business Process Management        |
| CDN     | Content Delivery Network           |
| CISO    | Chief Information Security Officer |
| CRM     | Customer Relationship Management   |
| DR      | Disaster Recovery                  |
| DMS     | Document Management System         |
| DMZ     | Demilitarized Zone                 |
| EA      | Enterprise Architecture            |
| ESB     | Enterprise Service Bus             |
| ETL     | Extract, Transform, Load           |

|         |                                              |
|---------|----------------------------------------------|
| FNOL    | First Notice of Loss                         |
| GDPR    | General Data Protection Regulation           |
| HSM     | Hardware Security Module                     |
| IAM     | Identity and Access Management               |
| KPI     | Key Performance Indicator                    |
| LAN     | Local Area Network                           |
| MTTR    | Mean Time to Recovery                        |
| NPS     | Net Promoter Score                           |
| PAS     | Policy Administration System                 |
| PCI-DSS | Payment Card Industry Data Security Standard |
| PoC     | Proof of Concept                             |
| RBAC    | Role-Based Access Control                    |
| RPO     | Recovery Point Objective                     |
| RTO     | Recovery Time Objective                      |
| SAN     | Storage Area Network                         |
| SLA     | Service Level Agreement                      |
| SOX     | Sarbanes-Oxley Act                           |
| STP     | Straight-Through Processing                  |
| VLAN    | Virtual Local Area Network                   |
| VPN     | Virtual Private Network                      |
| WAN     | Wide Area Network                            |

**Detailed Revision Notes for Version 2.1**

Section 2.3 was updated to reflect the new naming convention adopted by the Platform Engineering team. All references to 'microservice cluster' have been replaced with 'application workload group' per the Architecture Board decision of January 2025. Section 4.1 was expanded to include the results of the capacity planning exercise conducted in Q4 2024, which identified the need for additional compute resources in the Amsterdam primary data center.

Appendix B was added to include the RACI matrix requested by the PMO during the Q3 2024 quarterly review. The glossary was updated to include twelve new acronyms that were introduced since the previous version. Figure 3 was replaced with an updated diagram generated from the Sparx EA repository on March 28, 2025. The document classification was elevated from 'Internal' to 'Internal - Confidential' following the data classification review of February 2025.

Feedback from the peer review conducted by M. de Vries on March 10, 2025, addressed three items: (1) clarification of the boundary between the DMZ and the application tier, (2) addition of the backup retention policy reference, and (3) correction of the SLA target for the Claims Handling Service from 99.5% to 99.9% as agreed in the contract amendment of December 2024.

*DISCLAIMER: This document has been prepared for internal use by ArchiSurance B.V. The information contained herein is provided on an 'as-is' basis without warranties of any kind, either express or implied. ArchiSurance B.V. shall not be liable for any direct, indirect, incidental, or consequential damages arising from the use of this document. The views expressed herein are those of the Enterprise Architecture team and do not necessarily reflect the official position of the Board of Directors. All trademarks mentioned belong to their respective owners. This document may contain forward-looking statements that involve risks and uncertainties. Actual results may differ materially from those projected. Recipients are advised to seek independent professional advice before acting on any information contained in this document.*